

NILAVAN SRITHARAN

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PROFESSIONAL EXPERIENCE

Samsung Electronics

Software Development Engineer

Chennai, India

Oct 2022 - Dec 2023

- Developed the replacement feature, which reduced the time required for the customer to initiate and complete the replacement requests and increased the overall **customer satisfaction rate by 57%**.
- Optimised fulfillment service by upgrading to Node.js 18, refactored test cases, increased the **test success rate by 10%** and improved service deployment time and **memory usage efficiency by 15%**.
- Led migration of omnichannel functionalities to a microservice architecture, achieving an **80% improvement** in system response time and simplifying maintenance.
- Managed the insurance aspect of the service and worked closely with other teams for the implementation and integration of a new insurance provider plugin, debugging insurance-related issues, and creating scripts to automate unprocessed insurance claims, eliminating manual, time-consuming work.

Zoho Corp.

Project Trainee

Chennai, India

Mar 2022 - Jul 2022

- Designed a centralised school information management system (EmberJS, PostgreSQL) with quiz creation, exam scheduling, and academic records, reducing manual work and improving data accuracy by 80%.
- Built a file permission viewer in Java to extract ACLs of files/directories, enabling users to verify access rights 40% faster than manual checks.

SKILLS

Programming: Python, Java, JavaScript, SQL, R

Backend & Frameworks: Node.js, ReactJS, RestAPI

Databases & Tools: PostgreSQL, Redis, RabbitMQ, Jenkins

Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms, Unit Testing, Microservices

EDUCATION

University of Leeds

Master of Science in Data Science & Analytics

Leeds, United Kingdom

Expected Sept 2026

Relevant Modules: Statistical Theory & Methods, Machine Learning, Deep Learning, Business Analytics.

International Excellence Scholarship - Awarded 50% scholarship for MSc based on academic merit.

Anna University, St. Joseph's College of Engineering

Bachelor of Technology in Information Technology | CGPA: 8.47 / 10

Chennai, India

Jun 2018 - Apr 2022

Relevant Modules: OOP, Data Structures, DBMS, Algorithm Design

PERSONAL PROJECTS

Global Climate Attitudes Modelling (Python, Scikit-learn, Pandas)

[Github Link](#)

- Analysed 45,000+ UNDP survey responses from 73 countries, engineering 68 features to train classification models predicting climate commitment support. Logistic Regression achieved 82.8% accuracy, outperforming Random Forest.
- Applied K-Means clustering to identify 3 global country typologies, revealing climate attitudes (not demographics) as the primary predictor of support, with a distinct North-South divide in climate urgency.

Leeds Crime Patterns (Python, GeoPandas, Pandas)

[Github Link](#)

- Processed 100,000+ UK Police crime reports across West Yorkshire using geospatial joins with ONS ward boundaries, uncovering a statistically significant crime fingerprint in student-heavy wards, elevated burglary (6.9%) and vehicle crime (11.2%) vs city-wide average.
- Identified a recurring October spike in anti-social behaviour correlating with the university term start, demonstrating how temporal and demographic factors interact in urban crime distribution.

Flexi Book (Python, PostgreSQL, psycopg2, bcrypt)

[Github Link](#)

- Built a full CLI-based slot booking system from scratch with dual user/admin roles, layered architecture (handlers, models, services, utils), bcrypt authentication, and a PostgreSQL data layer handling booking logic, slot availability, and a gamified discount voucher system.